

	FIR VEHICLE TESTING	29/05/2023
FIR No :FIR/Testing-Outdoor/518100/2023/6035		
Part No/ Group :51790251R		
Subject :Performance evaluation of Signa 2823.T Cx RDE in comparison with Ashok Leyland 2820.T BSVI and Signa 2818.T BSVI base vehicle.		
Test Vehicle	Project Name :Signa 2823.T Cx BSVI RDE Model Name : Signa 2823.T Cx BSVI RDE	Enclosures : Annexure 1 to Annexure 7
	Chasis Number : PC03505022RJ00755 Engine Number : 12L63889392	
FIR Status :	Completed	

Role	Name	Signature
Prepared By	Debabrata Das	 29-May-2023
Checked By	Swapnil Jadhav	 30-May-2023
Approved By	Ajaykumar Jadhav	 30-May-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

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 FIR/Testing-Outdoor/518100/2023/6035		FIR VEHICLE TESTING		Date: 29/05/2023	
Part No / Group: 51790251R		Specification no :- " 259051790003"		Action by	Failure class
Rate of acceleration of Signa 2823.T Cx BSVI RDE vehicle is superior to AL 2820.T BSVI in 1st, 2nd & 3rd gears and inferior to AL 2820.T in 4th, 5th & 6th Gear.					
1.3 <u>Bathtub drivability:</u>					
➤ <u>Signa 2823.T RDE in comparison with Ashok Leyland 2820.T</u>					
Bathtub drivability of Signa 2823.T Cx BSVI RDE vehicle was found better as compared to Ashok Leyland 2820.T BSVI vehicle in 4th, 5th & 6th gear and meets PALS revised target.				--	--
➤ <u>Signa 2823.T BSVI RDE in comparison with Signa 2818.T BSVI</u>					
Bathtub drivability of Signa 2823.T Cx BSVI RDE vehicle was found superior to Signa 2818.T BSVI vehicle in all gears.				--	--
1.4 <u>Sustain and Restart Gradability:</u>					
➤ <u>Signa 2823.T Cx BSVI RDE in comparison with Ashok Leyland 2820.T BSVI</u>					
Sustain gradeability of Signa 2823.T Cx BSVI RDE was found to be 30.25 %. Signa 2823.T Cx BSVI RDE is 17.25% superior as compared to Ashok Leyland 2820.T BSVI vehicle.				--	--
Restart gradeability of Signa 2823.T Cx BSVI RDE was found to be 24.93%. Signa 2823.T Cx BSVI RDE is 16.93% superior as compared to Ashok Leyland 2820.T BSVI vehicle.				--	--
➤ <u>Signa 2823.T Cx BSVI RDE in comparison with Signa 2818.T BSVI</u>					
➤ Sustain gradeability of Signa 2823.T Cx BSVI RDE is 8.93% superior as compared to Signa 2818.T BSVI base vehicle.				--	--
➤ Restart gradeability of Signa 2823.T Cx BSVI RDE is 8.20% superior as compared to Signa 2818.T BSVI base vehicle.					
FIR Status				Page 2 of 5	
2.0 Result: Wide observation.					

 FIR/Testing-Outdoor/518100/2023/6035		FIR VEHICLE TESTING		Date: 29/05/2023		
Part No / Group: 51790251R		Specification no :- " 259051790003"			Action by	Failure class

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 2823.T Cx BSVI RDE (ERC – 1219)
Chassis No. – PC03505022RJ00755
Engine No. – 12L63889392

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan
Annexure 6: EOL

Input Doc Ref. : Testing-Outdoor/15076

DID No : NA

DDP No : NA

Work Order No. :- P.21.EA.501

Make :- Tata Motors Ltd

T I No. – 42070

Earlier Ref: -

- FIR/Testing-Outdoor/ 508500/2021/3980 Report on performance evaluation of Ashok Leyland 2820 HG BSVI vehicle in Comparison with Signa 2818.T Ashwamedh and Signa 2821.T 5L BSVI vehicle.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 2823.T Cx BSVI RDE vehicle was offered for performance evaluation, Ashok Leyland 2820.T BSVI was taken as core benchmark and Signa 2818.T BSVI vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

FIR Status

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 TATA FIR/Testing-Outdoor/518100/2023/6035		FIR VEHICLE TESTING	Date: 29/05/2023	
Part No / Group: 51790251R	Specification no :- " 259051790003"		Action by	Failure class

6.0 Testing

➤ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters		Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2	
			SIGNA 2818.T BSVI	AL 2820.T BSVI	SIGNA 2823.T RDE BSVI	
1	Vehicle Identification No. (ERC No./EJ No./Reg. No.)		ERC-1183	--	ERC-1219	
2	Vehicle GVW/GCW	kg	28000	28000	28000	
3	Engine Details (Make/cc/Norms/no. of Cyl.)		Cummins BS-VI Ashwamedh 5.6lit	H-Series CRS BSVI	Cummins BS-VI Ashwamedh 5.6lit	
	Engine Max Power		180 HP @ 2300rpm	200 hp @ 2400 rpm	220 HP	
	Engine Max Torque		850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	925Nm @ 1000-1300rpm	
	Engine Calibration Maturity Level		As per production released	As per production released	95% Cal Maturity(XWTPA19850)	
4	Regeneration Strategy (Time ase/Soot base)		Time Based (100 Hrs)	--	Time Based (100 Hrs)	
5	Gear Box (make/drive/FGR)		G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD	
	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36, 2-5.20, 3-3.04, 4-1.83, 5-1.32, 6-1.0	
6	RAR		4.88	5.57	4.88	
7	Tyre Details		All 295/90R 20 (Apollo Make)	F1 - Apollo 295/90R 20 16PR R1, R2 - Apollo 295/90R 20 16PR	All 295/90R 20 (Bridgestone Make)	
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900 rpm,1900 rpm	--	1900, 1900
		Vehicle Mass Threshold	kg	28000.00	--	28000
	VAM	Variable Acceleration 1, 2	kmph/sec	2.5,2.5	--	2.5,2.5
		Variable Acceleration Speed 1, 2	kmph	10, 40	--	10,40
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50, 50	--	50,50
	RSG	Maximum Accelerator Vehicle Speed	kmph	80.00	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%	--	105%
		Torque Limit Maximum Adjustment Factor	Economy	100%	--	115%
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
9	USFE / Torque Table Revision (1/2/3/n)		USFE-1	--	USFE-1	
	Fan Details (type/make/blades)		24" E-Viscous, Make - Behr , 11 no's blade	--	24" E-Viscous fan	
10	Base Cal Improvement (Cal Maturity%)		As per production released	As per production released	95% Cal Maturity	

Vehicle Photographs		NA	
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FIR/Testing-Outdoor/518100/2023/6035

FIR

VEHICLE TESTING

Date: 29/05/2023

Part No / Group: 51790251R

Specification no :- " 259051790003"

Action
byFailure
class**8.0 Torque table:-**

Signa 2823.T – Test vehicle

➤ Balance mode (TF – 115 %)

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

➤ Economy mode (TF – 105 %)

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

Signa 2818.T – Base vehicle

➤ Balance mode (TF – 100 %)

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	570	650	600	600	600	600	550	550	550	486	100	5
1.33	570	650	600	600	600	600	550	550	550	486	100	5
1.84	570	650	650	650	600	600	550	550	500	486	100	5
3.05	570	650	650	650	600	600	550	550	500	486	100	5
5.2	570	650	925	925	915	850	790	732	692	486	100	5
9.36	570	650	925	925	915	850	790	732	692	486	100	5

➤ Economy mode (TF – 100 %)

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	570	650	450	450	450	450	400	400	400	400	100	5
1.33	570	650	450	450	450	450	400	400	400	400	100	5
1.84	570	650	600	550	550	500	450	450	450	400	100	5
3.05	570	650	600	550	550	500	450	450	450	400	100	5
5.2	570	650	925	925	915	850	790	732	692	486	100	5
9.36	570	650	925	925	915	850	790	732	692	486	100	5

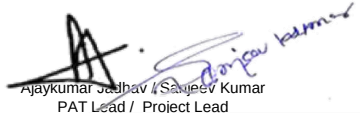
➤ **9.0 Observation:-**

Overall performance Signa 2823.T Cx BSVI RDE is better as compared to Ashok Leyland 2820.T BSVI and PALS position achieved "AL" against "AL".

10. Future testing:- As per further development.

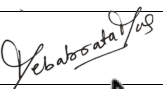
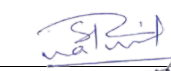
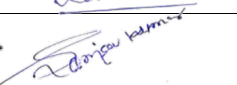
FIR Status

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Level 1 Attribute:					PANEL chart					VE CVBU/ PD/00													
PAT- Performance and Drivability					Specification: Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-950 DD 4.88 RAR 295/90R 20 Tyres 3 Mode FE Switch Torque Factor: E-115% ; B-105% EOL Parameters : LBSC: 1800,1800 RPM VAM: 2.5/2.5 kmph/s @ 10,40 kmph GDP: 45.45 kmph RSG: 80 kmph PTP: Disable					Panel Chart No.: VE CVBU/ M&HCV/ Signa 2823.T Cx BSVI RDE Ashwamedh/ PD/00													
Programme / MY : SIGNA 2823.T Cx 5.6L BSVI RDE			Gateway : DR3		Owner: Ajaykumar Jadhav		Project Name: SIGNA 2823.T 5.6L BSVI RDE(Cx Version)			Date: 29/05/2023													
Level 1 Summary:					Statement Of Intent (SOI) or DNA Description:																		
<table border="1"> <tr> <td rowspan="2">PALS</td> <td rowspan="2">Level 1:</td> <td>PALS</td> <td colspan="2">Status</td> </tr> <tr> <td>Current/ Gateway</td> <td>Job#1 Predicted</td> </tr> <tr> <td></td> <td></td> <td>AL</td> <td>AL</td> <td></td> </tr> </table>					PALS	Level 1:	PALS	Status		Current/ Gateway	Job#1 Predicted			AL	AL		Performance: Haulage application Fuel Economy: Haulage application						
PALS	Level 1:	PALS	Status																				
		Current/ Gateway	Job#1 Predicted																				
		AL	AL																				
Attribute Target Summary:																							
← Dev																							
	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Current Target w.r.t Ashok Leyland 3520.T BSVI	Status Current/Gateway SIGNA 2823.T Cx BSVI RDE	Base Vehicle Signa 2818.T BSVI	Competitor Data BS VI Core Benchmark Ashok Leyland 2820.T		Adjacent Benchmark	Remarks									
								Heavy mode	Heavy mode		--	--											
Specification								LBSC - 1900,1900,28000 Kg VAM- 2.5 Kmph/s, 2.5Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	LBSC - 1900,1900,28000 Kg VAM- 2.5 Kmph/s, 2.5Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph	--	--	--	--										
Vehicle GVW (Payload)						kg	28000	28000	28000	--	--	--	--										
Power (Max) @ rpm						HP	220 hp @ 2300 rpm	180 hp @ 2300 rpm	200 hp @ 2400 RPM	--	--	--	--										
Power to Weight ratio						HP / Ton	7.86	6.43	7.14	--	--	--	--										
Torque (Max) @ rpm						Nm	925 Nm @ 1000-1300 rpm	850 Nm @ 1000-1600 rpm	700 Nm at 1200-2000 rpm	--	--	--	--										
Torque to Weight ratio						Nm / Ton	33.04	30.36	25.00	--	--	--	--										
Tyres size						~	Bridgestone 295/90R20	Apollo 295/90 R20	Apollo JK Tyres 295/90 R20	--	--	--	--										
Gear Box Ratios						~	G950 DD, FGR -9.36	G950 DD, FGR -9.36	Eaton DD 6S Gearbox	--	--	--	--										
Final Drive Ratio						~	4.88	4.88	5.57	--	--	--	--										
Acceleration at rated load					AL		AL																
0-60 kmph						Sec	≤ 53.23	49.47	49.38	50.43	--	--	--										
0-1000 m						Sec	≤ 80.04	75.22	74.89	75.84	--	--	--										
Bathtub Drivability (Time) at Rated load					AL		AL																
3rd Gear						sec	≤ 4.3	4.27	3.05	3.24	--	--	--										
15 - 20 Kmph						sec	≤ 3.0	3.00	--	2.48	--	--	--										
20 - 25 Kmph						sec	≤ 3.0	2.93	--	2.62	--	--	--										
25 - 30 Kmph						sec	≤ 3.0																
4th Gear						sec	≤ 5.66	5.00	5.66	4.55	--	--	--										
20 - 25 Kmph						sec	≤ 4.33	3.85	3.97	4.33	--	--	--										
25 - 30 Kmph						sec	≤ 3.89	3.11	3.42	3.89	--	--	--										
30 - 35 Kmph						sec																	
5th Gear						sec	≤ 12.83	11.46	11.53	12.83	--	--	--										
30-40 kmph						sec	≤ 12.74	9.76	10.67	12.74	--	--	--										
40-50 kmph						sec	≤ 14.14	--	--	14.14	--	--	--										
50-60 kmph						sec																	
6th Gear						sec	≤ 27.76	26.35	27.76	17.44	--	--	--										
30-40 kmph						sec	≤ 17.68	15.49	15.79	17.68	--	--	--										
40-50 kmph						sec	≤ 17.77	12.45	14.89	17.77	--	--	--										
50-60 kmph						sec	≤ 20.12	15.66	--	20.12	--	--	--										
60-70 kmph						sec																	
Gradeability at rated load					AL		AL																
Sustain Gradeability						%	≥25.8	30.25	27.77	25.80	--	--	--										
Restart Gradeability						%	≥21.32	24.93	23.04	21.32	--	--	--										
Executive Summary:		1. Ashok Leyland 2820.T BSVI have been taken for target setting. 2. 2nd Gear(15-20 Kmph, 20-25 kmph, 25 - 30 Kmph), 4th Gear (20 - 25 Kmph) and 6th (30 - 40 Kmph) Gear Bath-tub driveability target is balanced as per target balancing sign off report 3. Signa 2823.T Cx BSVI RDE was having 95% engine calibration maturation. 4. Target has been taken at-par as per Cx variant approach																					
Tested by:					Verified and Approved by:																		
 Debabrata Das / Swapnil Jadhav PAT Engineer					 Ajaykumar Jadhav / Swapnil Kumar PAT Lead / Project Lead																		

TATA MOTORS LTD. ERC,PUNE		PAT OFFICE SHEET Performance and Drivability Vehicle-level Engine Calibration Features / End of Line OBDII and Diagnostics										Document Version: VE CVBU/PD/00										
		Department Name: Vehicle Engineering					Project Name: Signa 2823.T Atulya Cx BSVI RDE					Document Revision No.: VE CVBU/ M&HCV/ Signa 2823T Cx RDE Ashwamedh/ 00										
Date: 29/05/2023																						
Aggregate and test parameter											Base Vehicle		Core Benchmark		Test Vehicle							
Sr.No.	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Unit	Test Type	Classification (DRS)	Ref.Standard / Work Guideline	Simulation	Signa 2818.T 5.6L, Ashwamedh BSVI G950 DD 6S RAR 4.88	Ashok Leyland 2820.T Eaton 6 Speed DD RAR 5.57	Target	Signa 2823.T Cx RDE BSVI G950 DD RAR 4.88	Status (RYG)	Remarks	CoC Remark				
A	Performance Evaluation						Heading						LSJC- 1900,1900,28000 Kg VAM- 2.5 Km/hrh, 2.5Km/hrh, 10 km/hrh, 40km/hrh GDR- 30 km/hrh, 50 km/hrh R50 - 80km/hrh	--	--	LSJC- 1900,1900,28000 Kg VAM- 2.5 Km/hrh, 2.5Km/hrh, 10 km/hrh, 40km/hrh GDR- 30 km/hrh, 50 km/hrh R50 - 80km/hrh	--	1.Ashok Leyland 2823.T BSVI have been taken for target setting. 2.2nd Gear(15-20 Km/hrh, 20-25 Km/hrh, 25 - 30 Km/hrh), 4th Gear (20 - 25 Km/hrh) and 6th (20 - 40 Km/hrh) Gear both sub drivability target is balanced as per target balancing sign off report 3. Signa 2823.T Cx BSVI RDE was having 95% engine calibration maturity. 4.Target has been taken at par as per Cx variant approach				
1.1	Minimum speed						Heading						Heavy Mode		Heavy Mode	--	--					
1.1.1	1st gear						Km/hrh		Specification				3.2	2.7		2.8	--	--				
1.1.2	2nd gear						Km/hrh		Specification				5.8	5.3		4.9	--	--				
1.1.3	3rd gear						Km/hrh		Specification				10.1	8.8		8.8	--	--				
1.1.4	4th gear						Km/hrh		Specification				16.4	12.8		14.2	--	--				
1.1.5	5th gear						Km/hrh		Specification				21.9	18.9		19.3	--	--				
1.1.6	6th gear						Km/hrh		Specification				26.4	22.8		26.3	--	--				
1.1.7	7th gear						Km/hrh		Specification				NA	NA		NA	--	--				
1.1.8	8th gear						Km/hrh		Specification				NA	NA		NA	--	--				
1.2	Maximum speed						Heading						NA		NA	--	--					
1.2.1	1st gear						Km/hrh		Specification				10.9	12.4		7.9	--	--				
1.2.2	2nd gear						Km/hrh		Specification				19.8	21.1		14.1	--	--				
1.2.3	3rd gear						Km/hrh		Specification				33.7	34.7		26.1	--	--				
1.2.4	4th gear						Km/hrh		Specification				48.3	51.7		48.4	--	--				
1.2.5	5th gear						Km/hrh		Specification				49.6	71.4		61.2	--	--				
1.2.6	6th gear						Km/hrh		Specification				76.6(1900 RPM)	74.06(1700 RPM)		79.7	--	--				
1.2.7	7th gear						Km/hrh		Specification				NA	NA		NA	--	--				
1.2.8	8th gear						Km/hrh		Specification				NA	NA		NA	--	--				
1.2.9	Vehicle maximum speed						Km/hrh		Specification				80.0		80	80	0	--				
1.4.2	AC Off						Heading										--	--				
1.4.2.1	0-60 kmph (0-37 mph)						Sec		Standard				49.38	50.43	150.43	49.47	0	--				
1.4.2.4	0-1000 m						Sec		Target				74.89	75.84	575.84	75.72	0	--				
1.5	Grad ability						Heading										--	--				
1.5.2	AC Off						Heading										--	--				
1.5.2.1	Maximum (sustained) Gradability (forward)						%		Standard				27.77	25.8	326.8	30.25	0	--				
1.5.2.2	Max Resist Gradability (forward)						%		Standard				23.04	21.32	251.32	24.93	0	--				
1.6	Coast down (a+bV)						Heading										--	--				
1.6.2	Laden condition						Heading										--	--				
1.6.2.1	Power consumed at 40 kmph						KW						--	--	--	NA	--					
1.6.2.2	a						N						--	--	--	NA	--					
1.6.2.3	b						Nkmph ²						--	--	--	NA	--					
1.6.2.4	Coeff of rolling resistance						--						--	--	--	NA	--					
B	Drivability Evaluation						Heading										--	--				
1.3	WOT Drivability (AC Off)						Heading										--	--				
1.3.1	1st gear- 4-6 Km/hrh						Sec						1.02	0.84	--	0.89	--	--				
1.3.2	2nd gear- 8-12 Km/hrh						Sec															

Annexure 2

Aggregate and test parameter							Unit	Test Type	Classification (DR)	Ref.Standard / Work Guideline	Simulation	Base Vehicle Signa 2018.T 5.6L Aswamedh BSVI G950 DD 65 RAR 4.88	Core Benchmark Ashok Leyland 2020.T Eaton 6 Speed DD RAR 5.57	Target	Test Vehicle		Status (RYG)	Remarks	CoC Remark		
Sr.No.	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7									Signa 2023.T Cx RDE BSVI G950 DD RAR 4.88						
2.1	Vehicle Speed detection								Target				--	Pass	Pass	Pass	Green	--			
2.2	Engine Speed detection								Target				--	Pass	Pass	Pass	Green	--			
2.3	Gear Detection								Target				--	Pass	Pass	Pass	Green	--			
2.4	Driven Idle								Target				--	Pass	Pass	Pass	Green	--			
3.0	Engine derate verification								Heading				--	Pass	Pass	Pass	Green	--			
3.2	Limp home validation								Heading				--	Pass	Pass	Pass	Green	--			
3.3	Reduced torque fault validation								Heading				--	Pass	Pass	Pass	Green	--			
4.0	Evaluation of Take-off ability, Acceleration and Deceleration								Target				--	Pass	Pass	Pass	Green	--			
D	Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segregation and verification								Heading										--		
5.0	Vehicle Mass Estimation								Target				--	Pass	Pass	Pass	Green	--			
5.0	Road Slope Estimation								Target				--	Pass	Pass	Pass	Green	--			
5.0	Vehicle acceleration management (VAM tuning)								Target				--	Pass	Pass	Pass	Green	--			
6.0	Electronic vehicle speed Limiter (EVSL) / Load based speed control (LBSC)								Target				--	Pass	Pass	Pass	Green	--			
6.0	Vehicle speed limiter (Top and one-gear down)								Target				--	Pass	Pass	Pass	Green	--			
6.0	EVSL + VAM / LBSC+VAM								Target				--	Pass	Pass	Pass	Green	--			
7.0	Gear Down Protection (GDP)								Target				--	Pass	Pass	Pass	Green	--			
8.0	Idle shutdown								Target				--	Pass	Pass	Pass	Green	--			
8.0	Average fuel economy (AFE)								Target				--	Pass	Pass	Pass	Green	--			
10.0	IPE								Target				--	Pass	Pass	Pass	Green	--			
11.0	Engine Brake control								Target				--	Pass	Pass	Pass	Green	--			
12.0	Cruise control (CC)								Target				--	Pass	Pass	Pass	Green	--			
13.0	Power Take off (PTO)								Target				--	Pass	Pass	Pass	Green	--			
14.0	Remote PTO								Target				--	Pass	Pass	Pass	Green	--			
15.0	Road Speed governor								Target				--	Pass	Pass	Pass	Green	--			
16.0	Engine Protection shutdown								Target				--	Pass	Pass	Pass	Green	--			
17.0	Powertrain protection								Target				--	Pass	Pass	Pass	Green	--			
18.0	Trip-info								Target				--	Pass	Pass	Pass	Green	--			
E	Environmental Trials Verification								Heading										--		
1.0	Summer trials								Heading				--	Pass	Pass	Pass	Green	--			
1.4	Hot start								Target				--	Pass	Pass	Pass	Green	--			
1.5	Low idle governor calibration								Target				--	Pass	Pass	Pass	Green	--	Carry over from OBD I validation		
1.6	Driven Idle								Target				--	Pass	Pass	Pass	Green	--			
1.7	Subjective Drivability Assessment								Target				--	Pass	Pass	Pass	Green	--			
2.0	Winter trials								Heading				--	Pass	Pass	Pass	Green	--			
2.2	Cold Start (at Ambient as per Application requirement)								Target				--	Pass	Pass	Pass	Green	--	Carry over from OBD I validation		
2.3	Smoke level (visually) during free acceleration after start								Target				--	Pass	Pass	Pass	Green	--			
2.4	Subjective Drivability Assessment								Target				--	Pass	Pass	Pass	Green	--			
6.0	High altitude trials/ calibration								Heading				--	Pass	Pass	Pass	Green	--			
6.2	Low idle governor calibration						rpm	New Development	Target				--	Pass	Pass	Pass	Green	--			
6.9.1	% Deterioration in Drivability with base calibration						%		Target				--	Pass	Pass	Pass	Green	--			
6.10.1	Turbocharger duty cycle mapping								Target				--	Pass	Pass	Pass	Green	--	Carry over from OBD I validation		
6.10.1.1	Smoke during driving present or not								Target				--	Pass	Pass	Pass	Green	--			
6.10.1.2	Assessment of turbocharger Whistling noise							New Development	Target				--	Pass	Pass	Pass	Green	--			
6.11.1	Power calculation						%		Target				--	Pass	Pass	Pass	Green	--			
F	Onboard Diagnostics								Heading										--		
5.0	OBD Diagnostic - TML tool verification								Heading				--	Pass	Pass	Pass	Green	--			
7.0	Derate Strategies for Critical Faults								Heading				--	Pass	Pass	Pass	Green	--	Carry over from OBD I validation		
8.0	Verification of OBD interface for Comprehensive Component Monitors								Heading				--	Pass	Pass	Pass	Green	--			
Sign Off															Index for Status						
Responsibility							Name							Signature			Off target, no resolution				
Prepared by: PAT Engineer							Debabrata Das/Swapnil							 			Marginally off target, resolution planned/low risk				
Checked & Approved by: PAT Lead / Project Lead							Ajaykumar Jadhav / Sanjeev kumar							 			Met target, no concern				

PED evaluation of Signa 2823.T RDE BSVI in comparison with Ashok Leyland 2820 BSVI and Signa 2818.T Ashwamedh BSVI

PED| 29.04.23

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

BE BOLD

Taking calculated **risk** is key to making progress. We act with confidence and **agility** to accomplish our goals

OWN IT

Feeling and acting **empowered** is critical to drive results. We have an **Owner's Mind-set** and each of us takes full responsibility for the outcomes

SOLVE TOGETHER

Leveraging our collective genius while holding each other **accountable** helps us deliver the best. We **collaborate** proactively and transparently to achieve innovative solutions

BE EMPATHETIC

Embracing diversity makes us stronger for differences are opportunities to learn. We work with **passion to delight customers** and deliver greater success to our stakeholders

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Sustain and restart Gradeability

Vehicle Specification Comparison

Sr. No.	Parameters			Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
				SIGNA 2818.T BSVI	AL 2820.T BSVI	SIGNA 2823.T RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1183	--	ERC-1219
2	Vehicle GVW/GCW	kg		28000	28000	28000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins BS-VI Ashwamedh 5.6lit	H-Series CRS BSVI	Cummins BS-VI Ashwamedh 5.6lit
	Engine Max Power			180 HP @ 2300 rpm	200 hp @ 2400 rpm	220 HP
	Engine Max Torque			850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	925Nm @ 1000-1300rpm
	Engine Calibration Maturity Level			As per production released	As per production released	95% Cal Maturity(XWTPA19850)
4	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)	--	Time Based (100 Hrs)
5	Gear Box (make/drive/FGR)			G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36 , 2-5.20 , 3-3.04 , 4-1.83 , 5-1.32 , 6-1.0
6	RAR			4.88	5.57	4.88
7	Tyre Details			All 295/90R 20 (Apollo Make)	F1 - Apollo 295/90R 20 16PR R1, R2 - Apollo 295/90R 20 16PR	All 295/90R 20 (Bridgestone Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900 rpm,1900 rpm	--	1900, 1900
		Vehicle Mass Threshold	kg	28000.00	--	28000
	VAM	Variable Acceleration 1, 2	kmph/sec	2.5,2.5	--	2.5,2.5
		Variable Acceleration Speed 1, 2	kmph	10, 40	--	10,40
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50, 50	--	50,50
	RSG	Maximum Accelerator Vehicle Speed	kmph	80.00	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%	--	105%
		Torque Limit Maximum Adjustment Factor	Economy	100%	--	115%
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			USFE-1	--	USFE-1
9	Fan Details (type/make/blades)			24" E-Viscous, Make - Behr , 11 no's blade	--	24" E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	As per production released	95% Cal Maturity

Performance Data Summary

Annexure 3

Parameter	Unit	Target	PALS	Benchmark	Base Vehicle	Test Vehicle			RDE Status (%) w.r.t Benchmark	RDE Status (%) w.r.t Base	RDE Status w.r.t Target	PALS status
				Ashok Leyland 2820.T BSVI	Signa 2818.T BSVI HM	Signa 2823.T BSVI RDE HM	Signa 2823.T BSVI RDE BM	Signa 2823.T BSVI RDE LM				
Acceleration at Rated Load												
0 - 60 kmph	sec	≤ 50.43	AL	50.43	49.38	49.47	78.10	79.68	1.90%	-0.18%	1.90%	
0 - 1000 m	sec	≤ 75.84	AL	75.84	74.89	75.22	85.44	87.34	0.82%	-0.44%	0.82%	
Driveability(Rate of acceleration) at Rated Load												
1st Gear(4-6)	sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	0.84	1.02	0.89	0.96	1.05	Not Applicable/Data Generation	Not Applicable/Data Generation	Not Applicable/Data Generation	-
2nd Gear(8-12)	sec			2.32	1.92	1.96	2.05	1.92				
3rd Gear(12-20)	sec			6.04	4.86	4.64	5.36	5.50				
4th Gear(19-31)	sec			9.95	10.18	10.90	12.14	12.38				
5th Gear(25-42)	sec			21.22	20.32	21.14	25.48	26.46				
6th Gear(36-50)	sec			23.40	22.67	24.36	30.39	30.22				
1st Gear(4-6)	kmph/sec	Not Applicable/ Data Generation	Not Applicable/ Data Generation	2.38	1.96	2.25	2.08	1.90	Not Applicable/Data Generation	Not Applicable/Data Generation	Not Applicable/Data Generation	-
2nd Gear(8-12)	kmph/sec			1.72	2.08	2.04	1.95	2.08				
3rd Gear(12-20)	kmph/sec			1.32	1.65	1.72	1.49	1.45				
4th Gear(19-33)	kmph/sec			1.21	1.18	1.10	0.99	0.97				
5th Gear(25-42)	kmph/sec			0.80	0.84	0.80	0.67	0.64				
6th Gear(36-50)	kmph/sec			0.60	0.62	0.57	0.46	0.46				
Bath-Tub Driveability at Rated Load												
3rd Gear												
15 - 20 Kmph	Sec	≤ 4.3	AL	3.24	3.05	4.27	4.18	4.42	-31.79%	-40.00%	0.70%	
20 - 25 Kmph	Sec	≤ 3.0	AL	2.48	--	3.00	2.97	3.06	-21.05%	--	0.00%	
25 - 30 Kmph	Sec	≤ 3.0	AL	2.62	--	2.93	3.40	3.47	-11.83%	--	2.33%	
4th Gear												
20 - 25 Kmph	Sec	≤ 5.66	AL	4.55	5.66	5.00	5.55	5.65	-9.93%	11.63%	11.66%	
25 - 30 Kmph	Sec	≤ 4.33	AL	4.33	3.97	3.85	4.30	4.62	11.13%	3.07%	11.13%	
30 - 35 Kmph	Sec	≤ 3.89	AL	3.89	3.42	3.11	4.71	5.20	20.10%	9.12%	20.10%	
35 - 40	Sec	--	--									
5th Gear												
30-40 kmph	Sec	≤ 12.83	AL	12.83	11.53	11.46	14.31	14.74	10.71%	0.64%	10.71%	
40-50 kmph	Sec	≤ 12.74	AL	12.74	10.67	9.76	--	--	23.41%	8.55%	23.41%	
50-60 kmph	Sec	≤ 14.14	AL	14.14	--	--	--	--	--	--	--	
6th Gear												
30-40 kmph	Sec	≤ 27.76	AL	17.44	27.76	26.35	26.38	26.42	-51.08%	5.09%	5.08%	
40-50 kmph	Sec	≤ 17.68	AL	17.68	15.79	15.49	19.14	19.98	12.39%	1.90%	12.39%	
50-60 kmph	Sec	≤ 17.77	AL	17.77	14.89	12.45	--	--	29.96%	16.41%	29.96%	
60-70 kmph	Sec	≤ 20.12	AL	20.12		15.66	--	--			22.17%	
Sustain and Restart Gradeability												
Sustained Gradeability	%	≥25.8	AL	25.8	27.77	30.25	--	--	17.25%	8.93%	17.25%	
Restart Gradeability	%	≥21.32	AL	21.32	23.04	24.93	--	--	16.93%	8.20%	16.93%	

Enclosed with FIP Testing Outdoor 518100/2023/6035

Signa 2823.T Cx BSVI RDE Performance. PALS position achieved "AL" against "AL"

Acceleration

- Acceleration of Signa 2823.T Cx BSVI RDE was found to be **better as compared to target** in 0-60 kmph speed and 0-1000 meters band

Bath tub Drivability

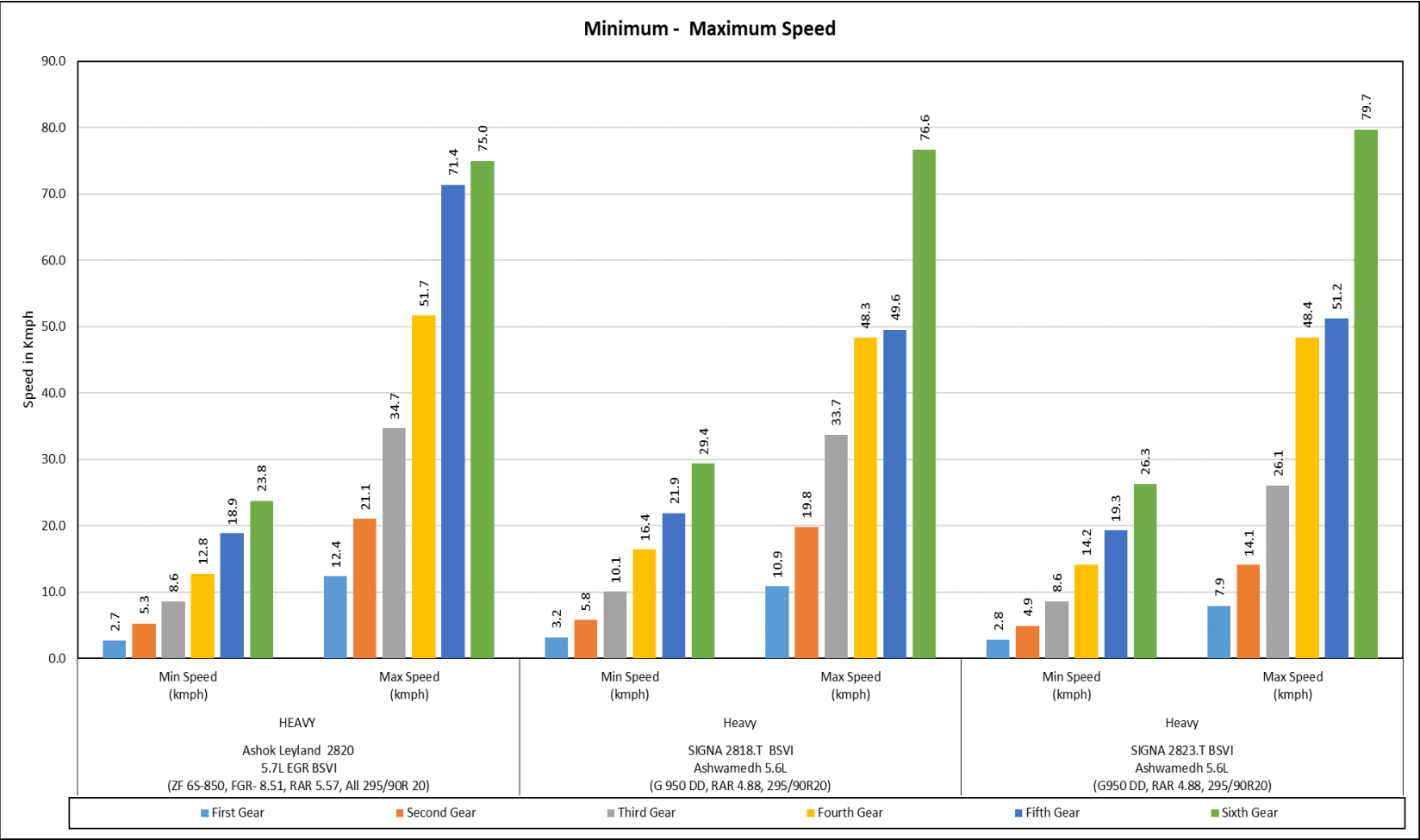
- Drivability of Signa 2823.T Cx BSVI RDE was found to be **better as compared to target** in all the speed bands

Sustain & Restart Gradeability

- Restart Gradability of 2823.T Cx RDE BSVI was **found better** than target.
- Sustain Gradability of 2823.T Cx RDE BSVI was **found better** that target

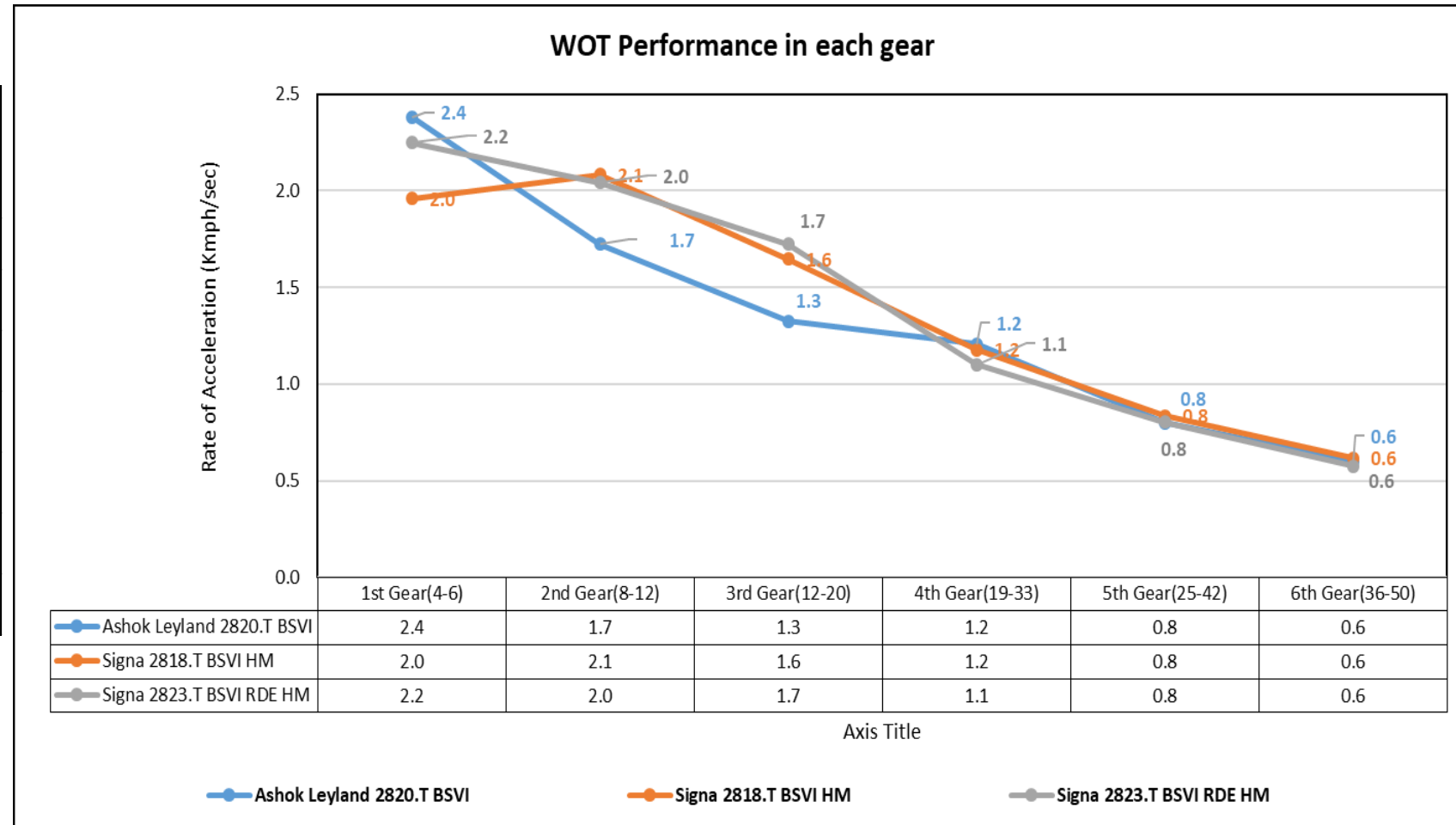
Minimum & Maximum Gear Wise speed Comparison

GEAR	Ashok Leyland 2820 5.7L EGR BSVI (ZF 6S-850, FGR- 8.51, RAR 5.57, All 295/90R 20)		SIGNA 2818.T BSVI Ashwamedh 5.6L (G 950 DD, RAR 4.88, 295/90R20)		SIGNA 2823.T BSVI Ashwamedh 5.6L (G950 DD, RAR 4.88, 295/90R20)	
	HEAVY		Heavy		Heavy	
	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)	Min Speed (kmph)	Max Speed (kmph)
1	2.7	12.4	3.2	10.9	2.8	7.9
2	5.3	21.1	5.8	19.8	4.9	14.1
3	8.6	34.7	10.1	33.7	8.6	26.1
4	12.8	51.7	16.4	48.3	14.2	48.4
5	18.9	71.4	21.9	49.6	19.3	51.2
6	23.8	75.0	29.4	76.6	26.3	79.7

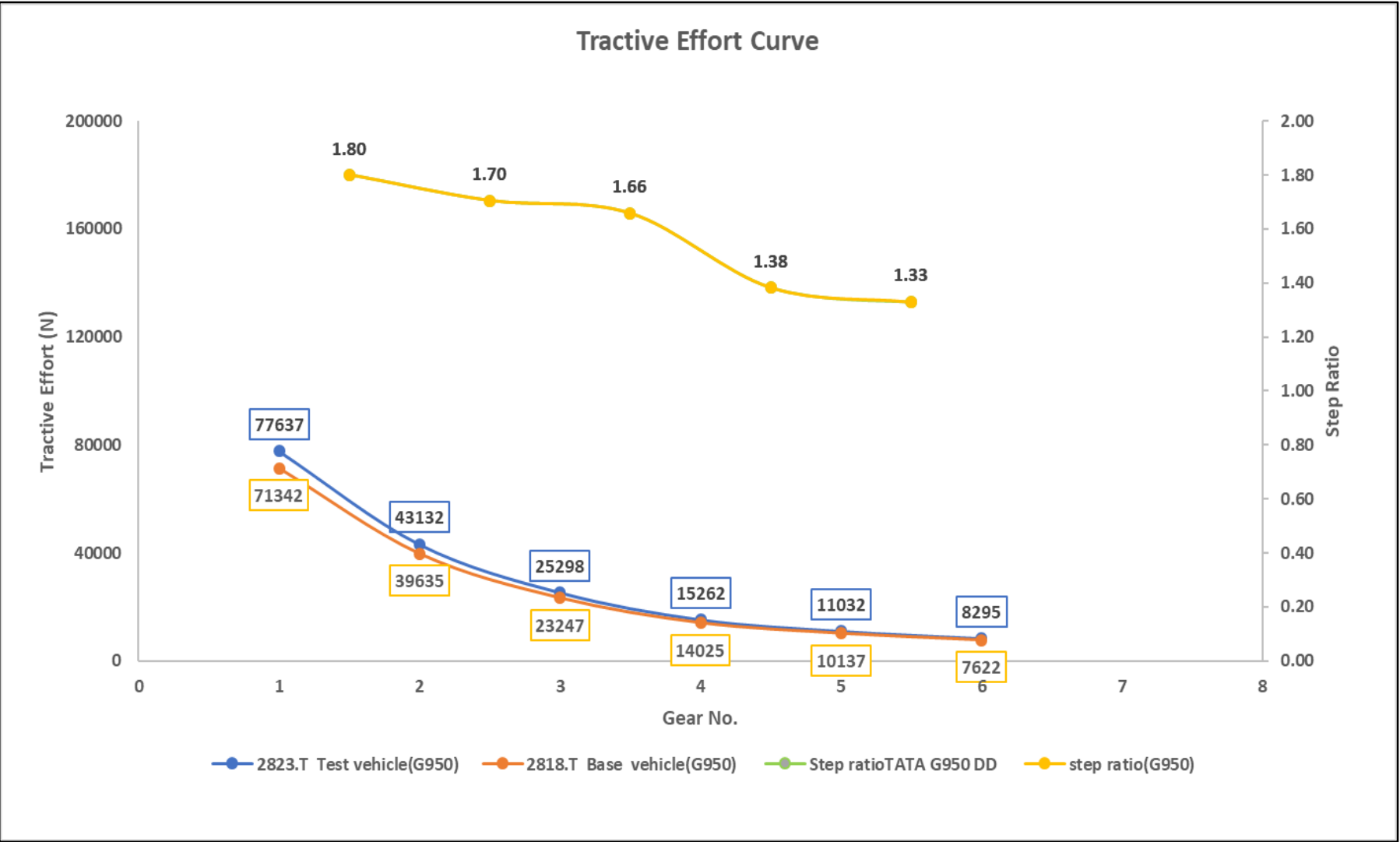


Wide open throttle Performance

Gear	Ashok Leyland 2820.T BSVI	Signa 2818.T BSVI HM	Signa 2823.T BSVI RDE HM
	Kmph/sec	Kmph/sec	Kmph/sec
1st Gear(4-6)	2.4	2.0	2.2
2nd Gear(8-12)	1.7	2.1	2.0
3rd Gear(12-20)	1.3	1.6	1.7
4th Gear(19-33)	1.2	1.2	1.1
5th Gear(25-42)	0.8	0.8	0.8
6th Gear(36-50)	0.6	0.6	0.6



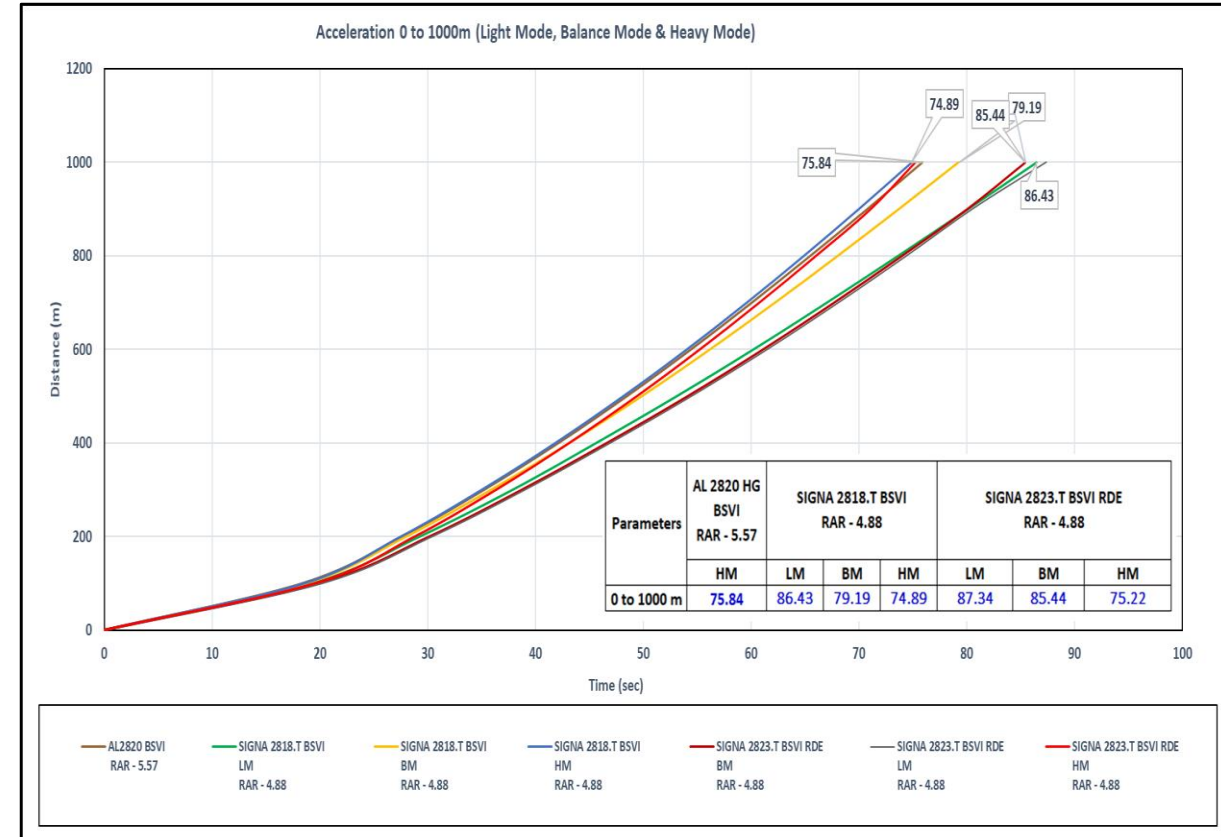
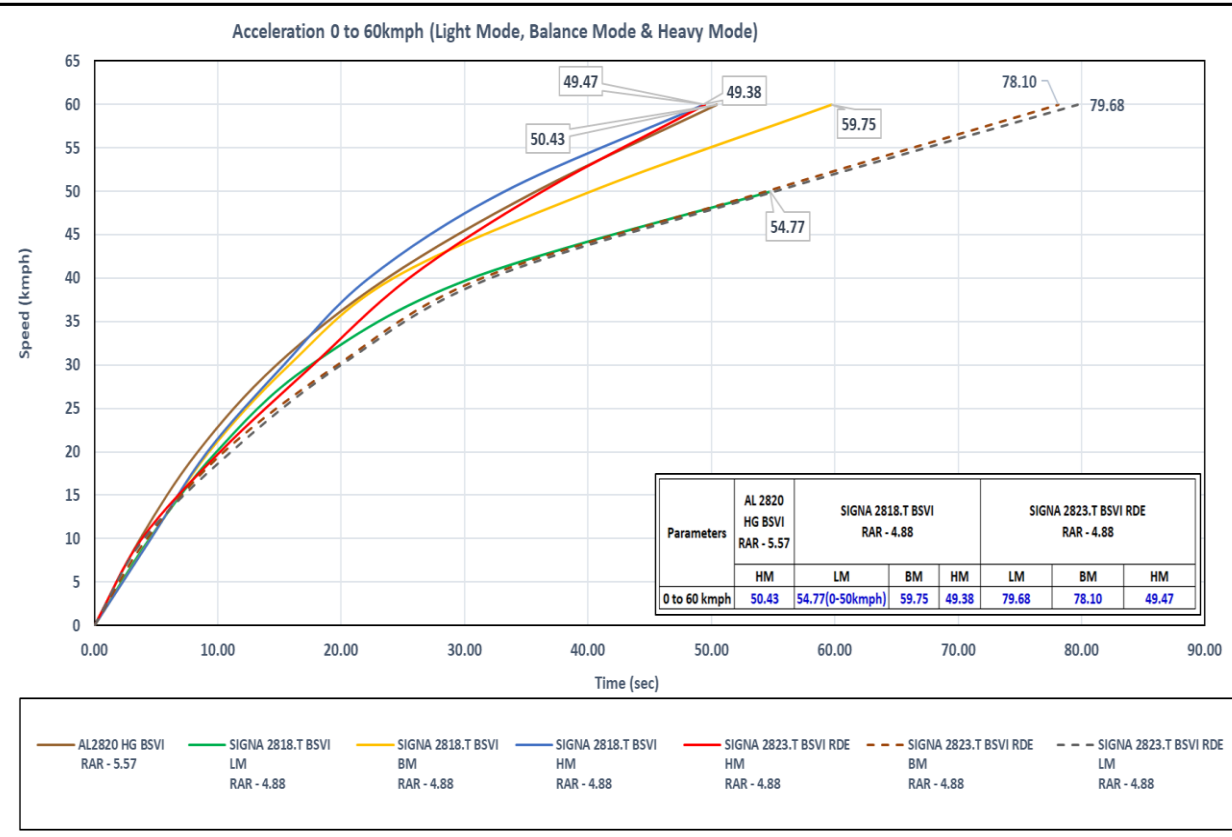
Tractive effort



Remarks:

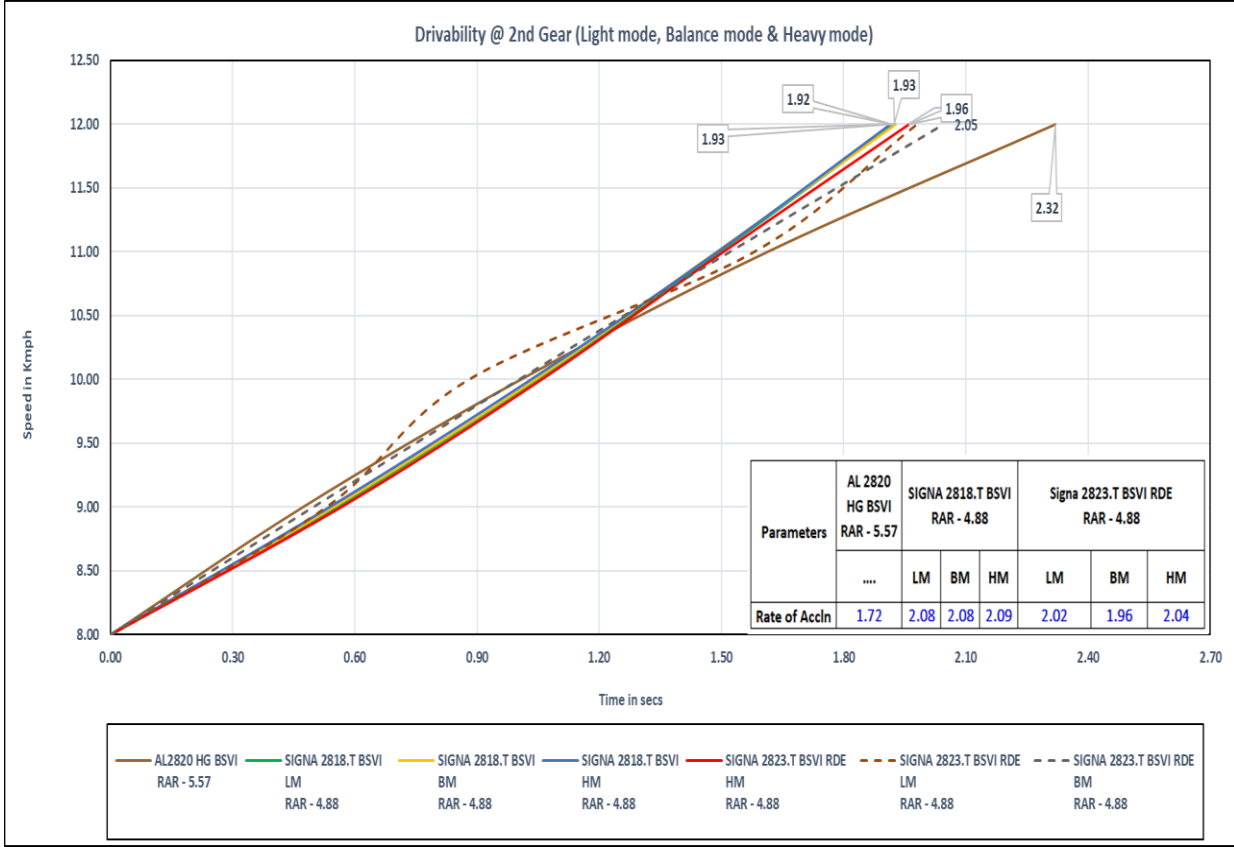
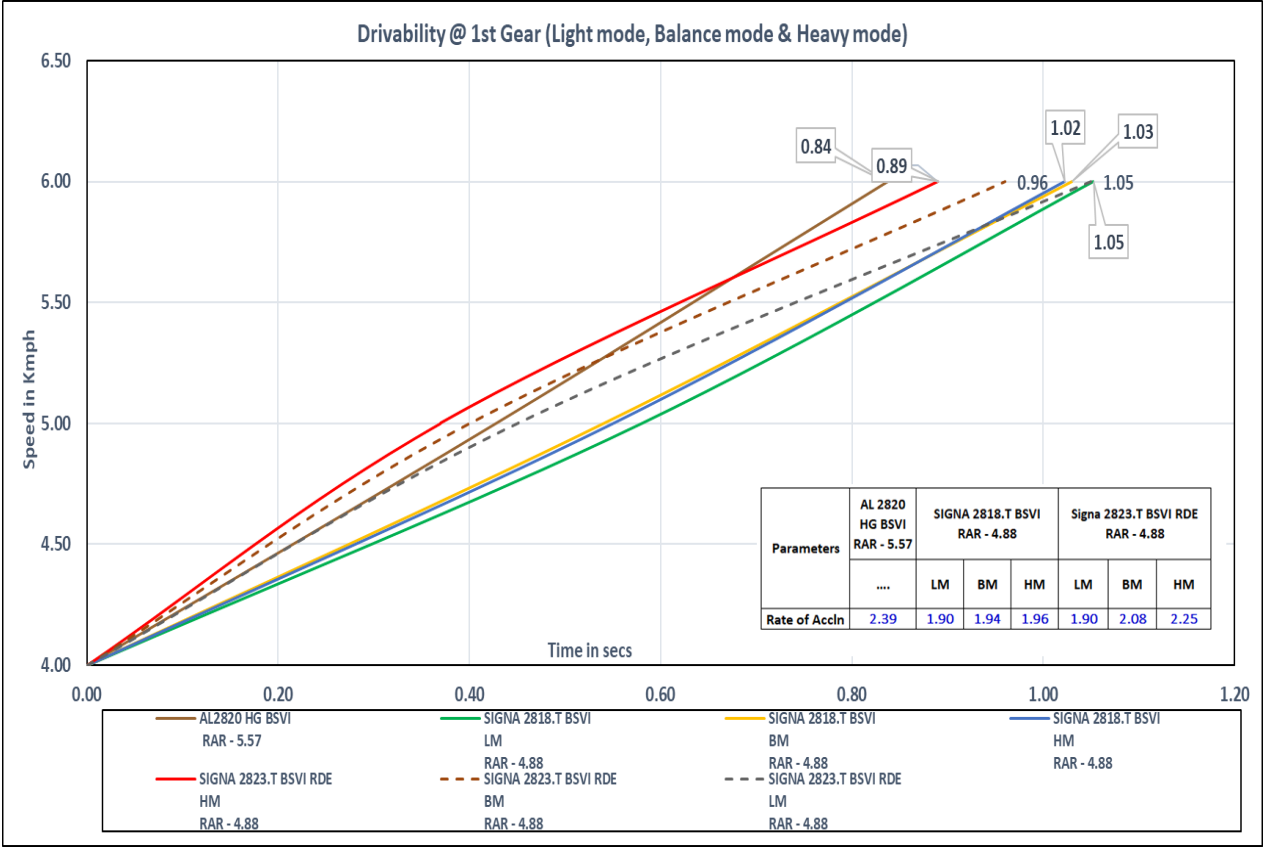
- Tractive effort of Signa 2823.T RDE BSVI is Superior than Base Vehicle vehicle in all gear.

Acceleration Performance – Through Gear



- ❖ In 0 – 60 Kmph, Through gear acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI
- ❖ In 0 – 1000 mts, Through gear acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** AL 2820.T BSVI
- ❖ In 0 – 60 Kmph, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** with Signa 2818.T BSVI
- ❖ In 0 – 1000 mts, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** Signa 2818.T BSVI

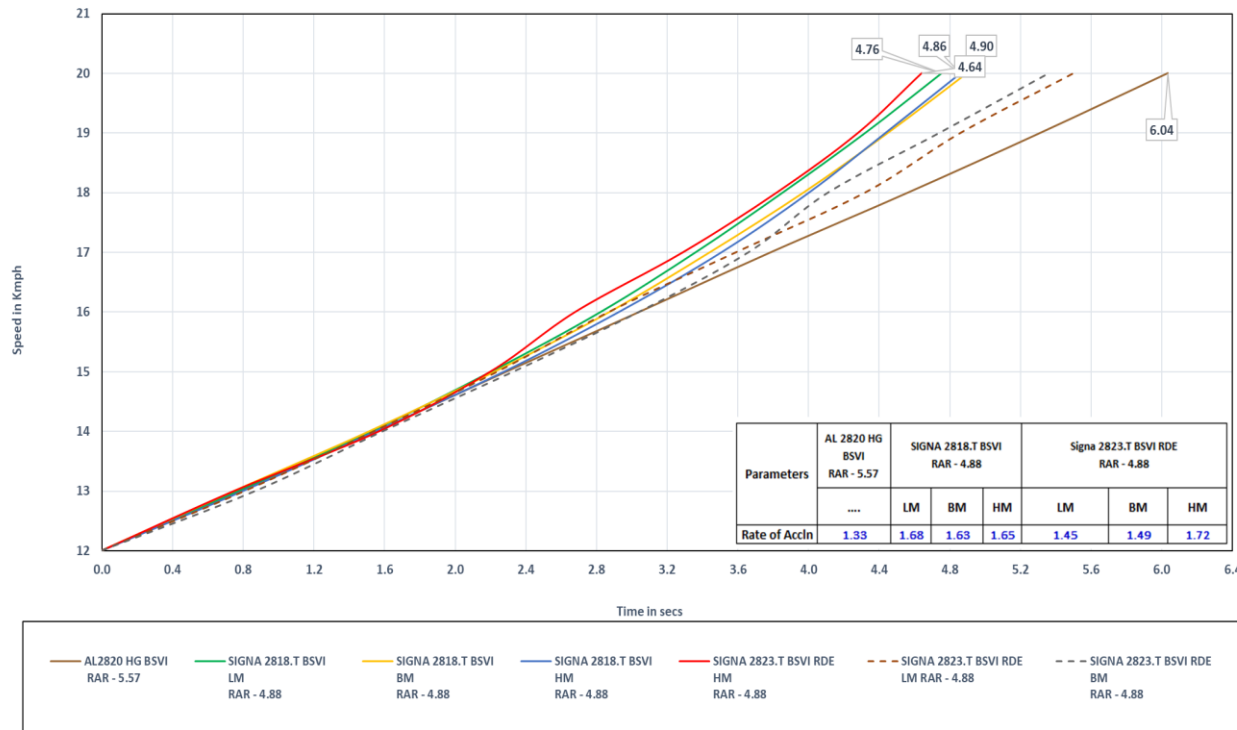
WOT Drivability @ 1st & 2nd Gear



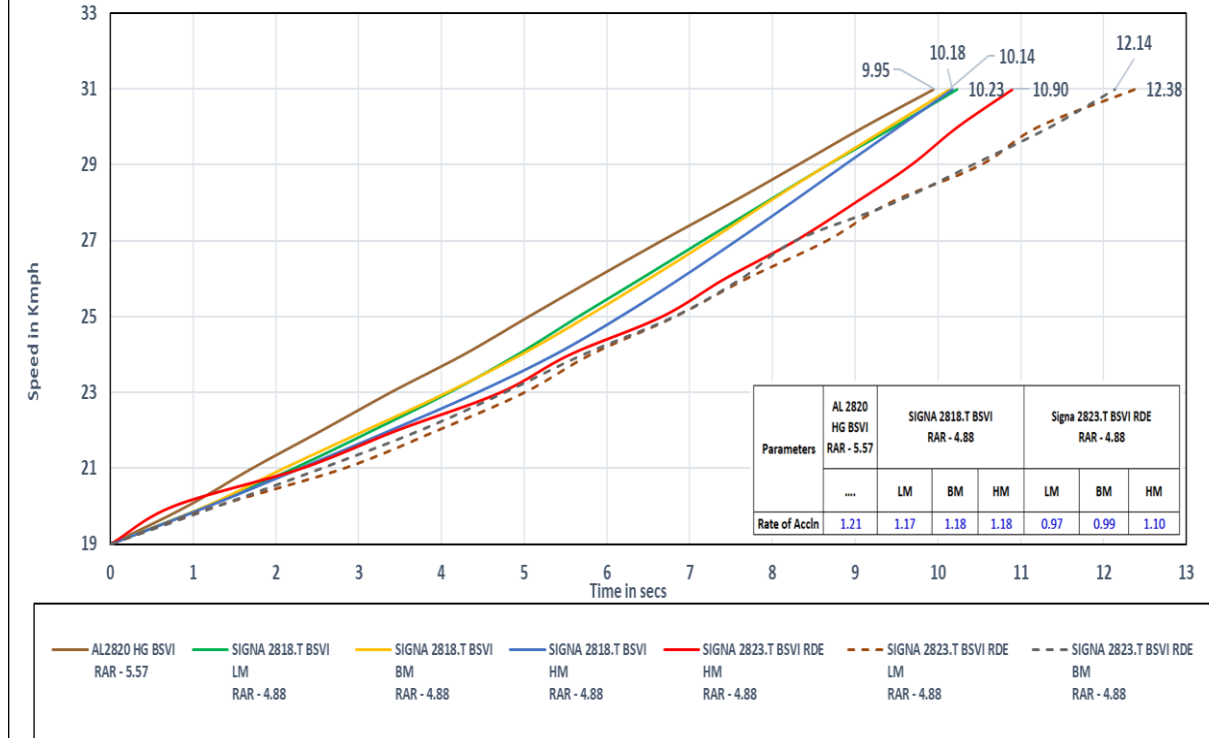
- ❖ In First gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI
- ❖ In Second gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI
- ❖ In First gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with Signa 2818.T BSVI
- ❖ In Second gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with Signa 2818.T BSVI

WOT Drivability @ 3rd & 4th Gear

Drivability @ 3rd Gear (Light mode, Balance mode & Heavy mode)



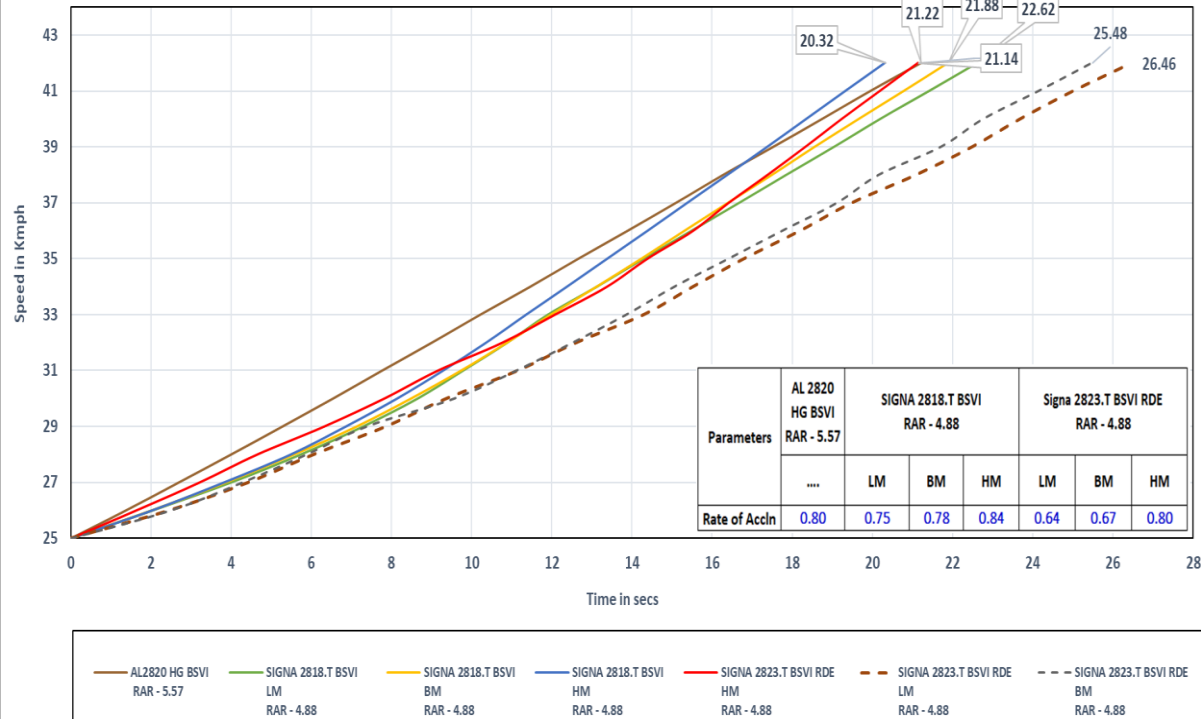
Drivability @ 4th Gear (Light mode, Balance mode & Heavy mode)



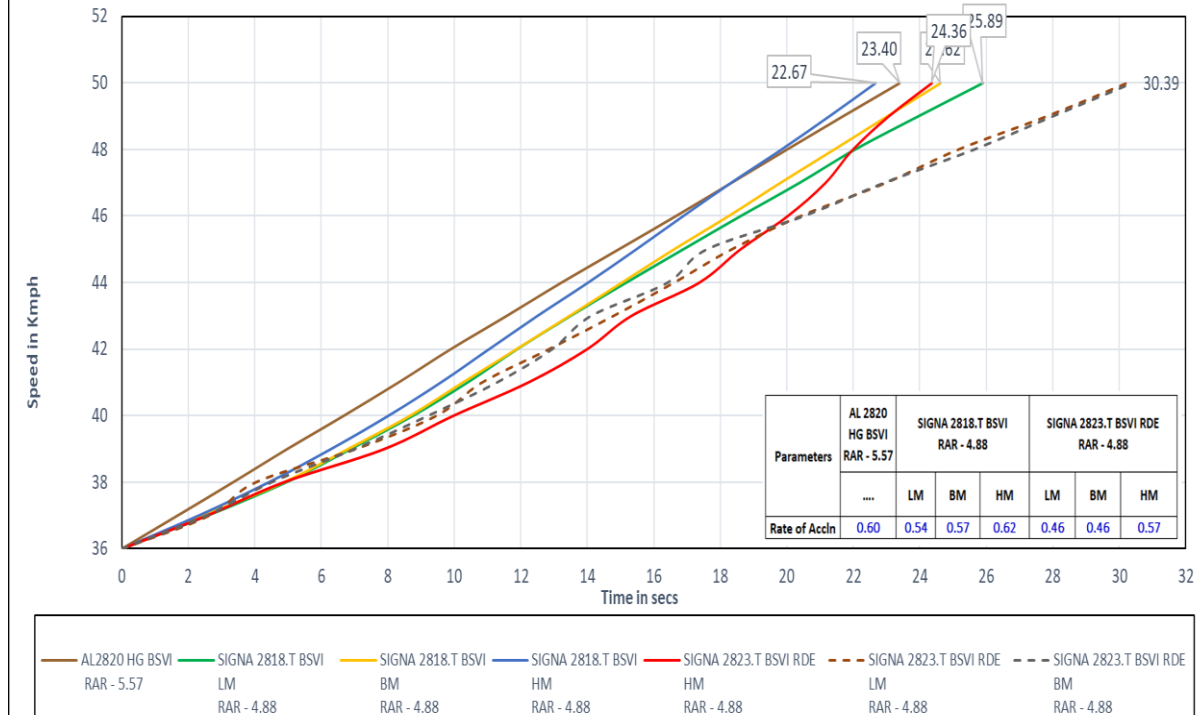
- ❖ In third gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI
- ❖ In fourth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** with AL 2820.T BSVI
- ❖ In third gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with Signa 2818.T BSVI
- ❖ In fourth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** with Signa 2818.T BSVI

WOT Drivability @ 5th & 6th Gear

Drivability @ 5th Gear (Light mode, Balance mode & Heavy mode)



Drivability @ 6th Gear (Light mode, Balance mode & Heavy mode)

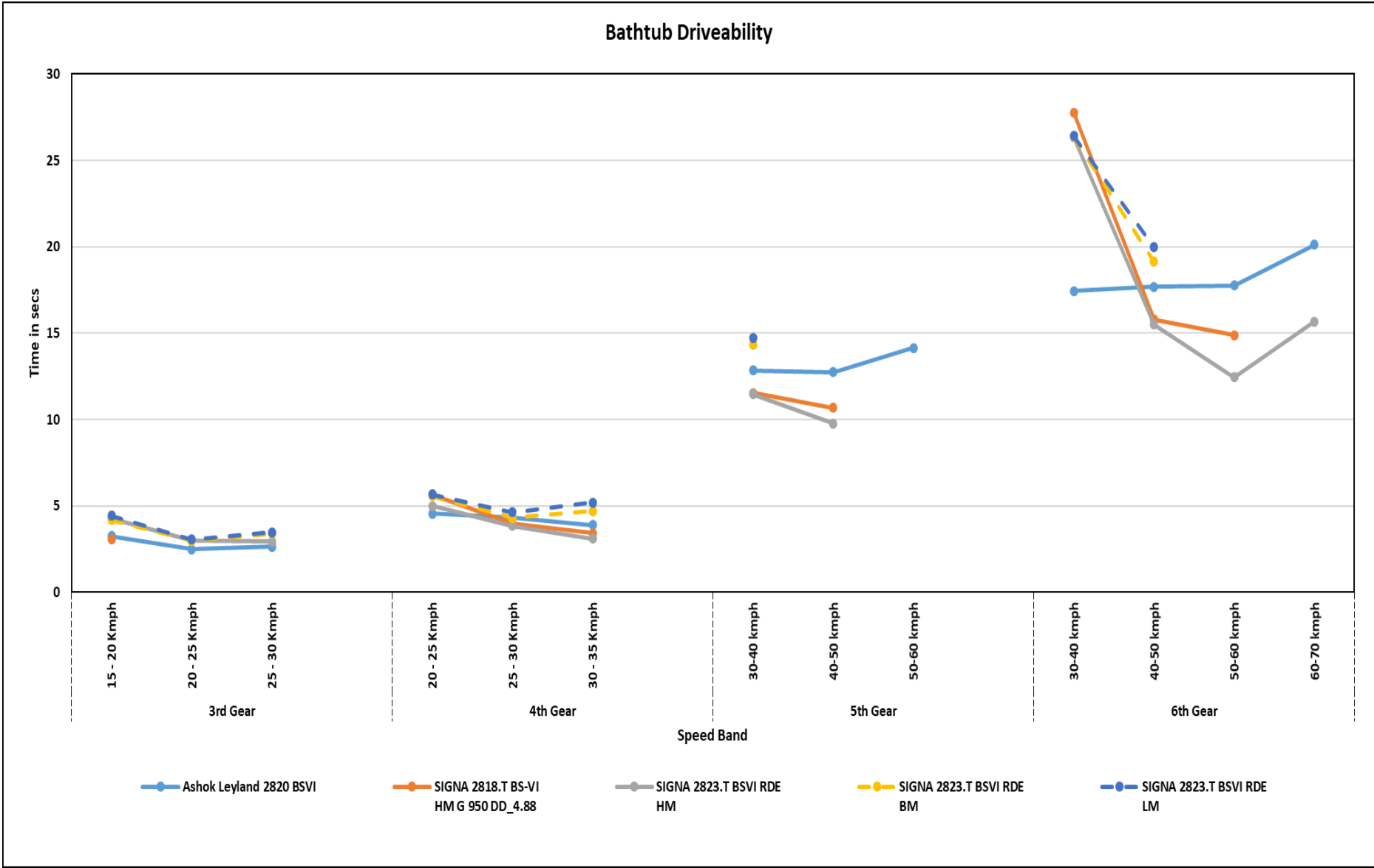


- ❖ In Fifth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is at par** with AL 2820.T BSVI
- ❖ In Sixth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** with AL 2820.T BSVI
- ❖ In Fifth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is better as compared** with Signa 2818.T BSVI
- ❖ In Sixth gear, WOT acceleration performance of Signa 2823.T Cx RDE vehicle **is inferior as compared** with Signa 2818.T BSVI

Bathtub Drivability Report

Gear	Speed Band	Target	Ashok Leyland 2820 BSVI	SIGNA 2818.T BS-VI HM G 950 DD_4.88	SIGNA 2823.T BSVI RDE HM	SIGNA 2823.T BSVI RDE BM	SIGNA 2823.T BSVI RDE LM
3rd Gear	15 - 20 Kmph	≤ 4.3	3.24	3.05	4.3	4.2	4.4
	20 - 25 Kmph	≤ 3.0	2.48		3.0	3.0	3.1
	25 - 30 Kmph	≤ 3.0	2.62		2.9	3.4	3.5
4th Gear	20 - 25 Kmph	≤ 5.66	4.55	5.66	5.0	5.6	5.7
	25 - 30 Kmph	≤ 4.33	4.33	3.97	3.8	4.3	4.6
	30 - 35 Kmph	≤ 3.89	3.89	3.42	3.1	4.7	5.2
5th Gear	30-40 kmph	≤ 12.83	12.83	11.53	11.46	14.31	14.74
	40-50 kmph	≤ 12.74	12.74	10.67	9.76		
	50-60 kmph	≤ 14.14	14.14				
6th Gear	30-40 kmph	≤ 27.76	17.44	27.76	26.35	26.38	26.4
	40-50 kmph	≤ 17.68	17.68	15.79	15.49	19.1	20.0
	50-60 kmph	≤ 17.77	17.77	14.89	12.45		
	60-70 kmph	≤ 20.12	20.12		15.66		

Bathtub Drivability Comparison with Benchmark & Base Vehicles

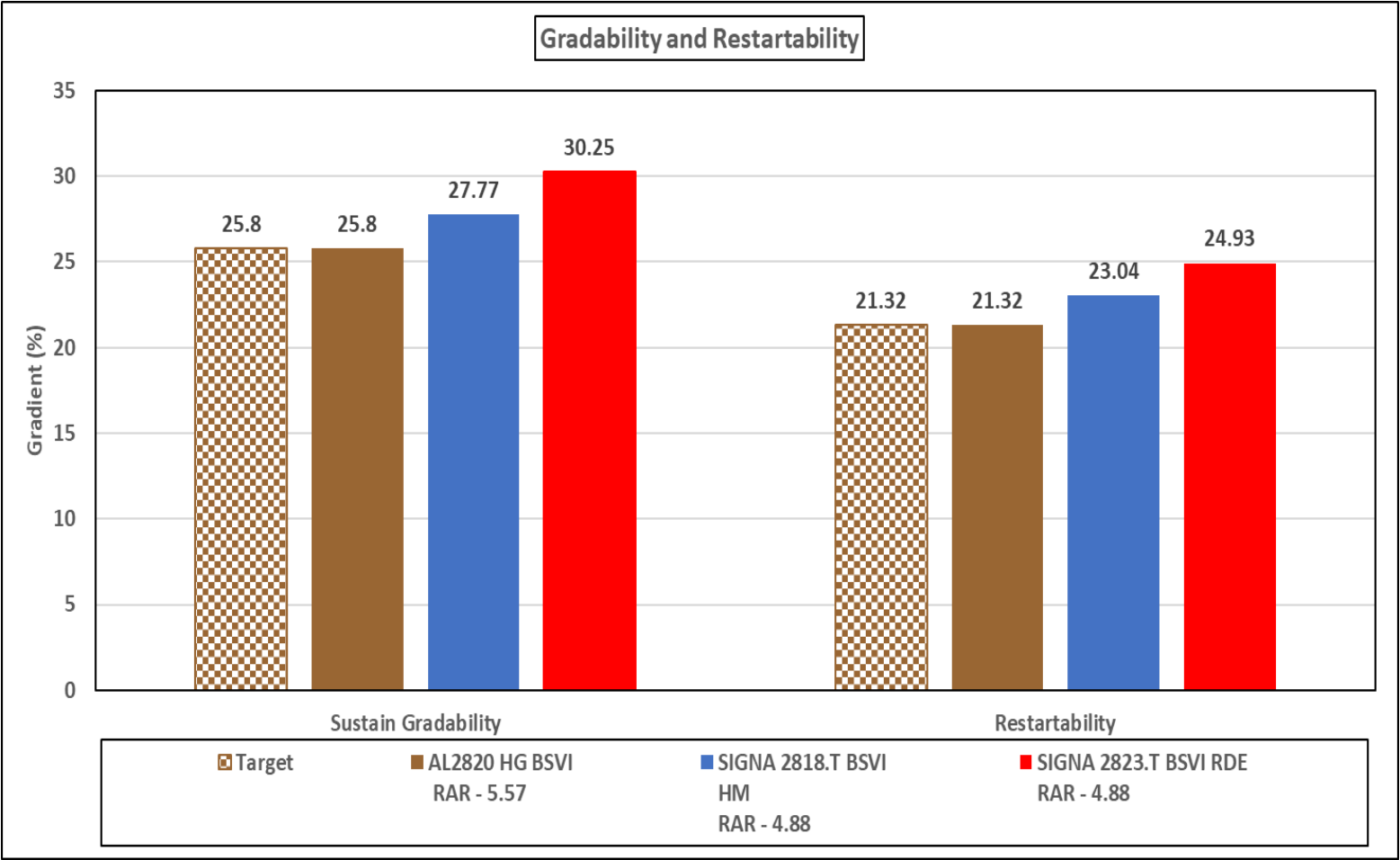


Remarks:

Vehicle is meeting PALS target in all gears

Sustained and Restart Gradeability

Test	Sustained & Restart Gradeability			
	Target	AL2820 HG BSVI RAR - 5.57	SIGNA 2818.T BSVI HM RAR - 4.88	SIGNA 2823.T BSVI RDE RAR - 4.88
Sustain Gradability	25.8	25.8	27.77	30.25
Restartability	21.32	21.32	23.04	24.93



Thank You!

TEST REQUISITION

Ref.		Date:	30/03/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 2823.T BSVI				
2	Part Number	51815248R,				
3	Drawing number	518100000005,	Mod status	C,2,	Mod Date	07/05/2023
4	Vehicle Model / Project	SIGNA 2825.T RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☒ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	RIDE AND HANDLING				
13	Remarks					
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 29/05/2023			
	VEHICLE TESTING DEPARTMENT				
Description: TATA SIGNA 2823.T BSVI 48WB; G950 GB; 4.88RAR; 295/90R20 TYRES		Test Item No: 42070			
		New	☒	Moderate	☒
		Major	☒	Minor	☒

Part No: 51815248R,	Modification No: C,2,	Enclosures:
Drawing No: 518100000005,		
Make: NA	WBS Number: P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	Copies to :
Vehicle Type: SIGNA 2825.T RDe BSVI	No. of Samples: 1	
Test Requested by:	VEHICLE-I	Equipments:
Test Requisition Reference:	Testing-Outdoor/15033	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component: b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Signa 2823.T PD trials	As per PED work guideline	As per PALS	1	Debabrata Das	Beta	Nil

REMARKS: Test plan for DVP execution	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: Yes		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY

VC No.					VC : 51815156R	VC : 51815256R
Vehicle Model					Signa 2823.T.AC	Signa 2823.T. Blower
Parameter	VEPS Name	DEFAULT VALUE *	MIN/MAX *	UNITS	Signa 2823.T Achwamedh (G-950 DD [4.88RAR [295/90R20]	Signa 2823.T Achwamedh (G-950 DD [4.88RAR [295/90R20]
Fuel Economy						
Gear Down Protection (GDP)						
Gear Down Protection Enable	20061	N		Y = Enable N = Disable	Y	Y
Gear Down Protection Heavy Load Vehicle Speed	11440	28	Min: 16 Max: (10000)	mph	45 kmph (27.96 mph)	50 kmph (31.25 mph)
Gear Down Protection Light Load Vehicle Speed	11430	28	Min: 19 Max: 56	mph	45 kmph (27.96 mph)	50 kmph (31.25 mph)
Idle Speed Control						
Adjustable Low Idle Speed Enable (Read Only)	50512	Y		49	Y	Y
Low Idle Speed	10030	750	Min: 600 Max: 800	rpm	650	650
Idle Shutdown						
Idle Shutdown Enable	20051	N		Y = Enable N = Disable	Y	Y
Idle Shutdown Timer	10170	60.0	Min: 0.5 Max: 1440.0	Minutes	5	5
Idle Shutdown in PTO Enable	20071	N		Y = Enable N = Disable	N	N
Idle Shutdown Manual Override Enable	20081	N		Y = Enable N = Disable	Y	Y
Idle Shutdown service brake switch interaction	50057	N		Y = Enable N = Disable	Y	Y
Load-Based Speed Control						
Load Based Speed Control Enable	50016	N		Y = Enable N = Disable	Y	Y
Load Based Speed Control Low Engine Speed Breakpoint	50027	1800	Min: 1200 Max: 2100	rpm	1800	1900
Load Based Speed Control High Engine Speed Breakpoint	50028	1800	Min: 1500 Max: 2500	rpm	1800	1900
Load Based Speed Control Vehicle Mass Threshold Enable (Read Only)	50446	Y		Y = Enable N = Disable	Y	Y
Load Based Speed Control Vehicle Mass Threshold	50447	68343	Min: 22046 Max: 220460	lb	28000 kg (61729.43 lbs)	28000 kg (61729.43 lbs)
Road Speed Governor						
Road Speed Governor Enable (Read Only)	99001	Y		Y = Enable N = Disable	Y	Y
Maximum Accelerator Vehicle Speed	10000	56	Min: 12 Max: (13590)	mph	80 kmph (49.709 mph)	80 kmph (49.709 mph)
User Adjustable Fuel Economy						
User Adjustable Fuel Economy Enable	50437	N		Y = Enable	Y	Y
				N = Disable		
				Y = Enable		
				N = Disable		
UAFE Max Road Speed Enable	50475	Y			Y	Y
UAFE Max Road Speed Limit	50476	36	Min: 12 Max: 56	mph	79.9 kmph (49.647 mph)	79.9 kmph (49.647 mph)
UAFE Torque Limit Enable	50486	N		Y = Enable N = Disable	Y	Y
UAFE Mid Torque Limit Adjust Factor	50487	100	Min: 80 Max: 200	%	125	125
UAFE Max Torque Limit Adjust factor	50488	100	Min: 80 Max: 200	%	115	115
Accelerator Control						
Variable Rate Vehicle Acceleration Management	50040	N		Y = Enable N = Disable	Y	Y
Variable Acceleration 1	50041	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration 2	50042	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration Speed 1	50043	6	Min: 0 Max: 124	mph	20 kmph (12.43 mph)	20 kmph (12.43 mph)
Variable Acceleration Speed 2	50044	37	Min: 0 Max: 124	mph	60 kmph (37.28 mph)	60 kmph (37.28 mph)
Engine Brakes						
Engine Brake Control Enable	23021	N		Y = Enable N = Disable	Y	Y
Engine Brake Type	12790	0		0 = Exhaust Brake 2 = Engine Brake	2	2
Engine Brake Minimum Vehicle Speed	11540	0	Min: 0 Max: 35	mph	0	0
Engine Brake Delay Time	11542	0.0	Min: 0.0 Max: 4095.9	Seconds	0	0
Engine Brake Service Brake Activation Enable	21251	N		Y = Enable N = Disable	N	N
Engine Brake Clutch Pedal Activation Enable	21253	N		Y = Enable N = Disable	Y	Y
Cruise Control and Engine Brake Interaction Enable	20601	N		Y = Enable N = Disable	N	N
Start Maximum Engine Braking in Cruise Control	12410	3.1	Min: 0.0 Max: 6.2	mph	--	--
Start Minimum Engine Braking in Cruise Control	12420	2.0	Min: 0.0 Max: 6.2	mph	--	--
Accelerator Base						
Clutch Switch	10790	0		0 = Not Installed 1 = Installed	1	1
Starter Lockout Enable	24000	N		Y = Enable N = Disable	Y	Y
Starter Lockout Relay Type	22161	0		0 = Normally Closed 1 = Normally Open	1	1
Transmission Setup						
Transmission Top Gear Ratio	10300	1.000	Min: 0.000 Max: 64.000	Ratio	1	1
Gear Down Transmission Ratio	11340	1.330	Min: 0.000 Max: 64.000	Ratio	1.33	1.33
Transmission Type	11410	0		0 = Manual ; 1 = Automatic ; 2 = Fully Automated with Clutch ; 3 = Partially Automated 4 = Fully Automated without Clutch	0	0
Vehicle Speed Input						
Rear Axle Ratio	10050	5.120	Min: 1.000 Max: 15.996		4.88	4.88
Tire Size	10060	500	Min: 301 Max: 1287	Rev/Mile	295 rev/km (475 rev/mile)	295 rev/km (475 rev/mile)
Number of Transmission Tailshaft Gear Teeth	10070	4	Min: 1 Max: 64		4	4
Vehicle Speed Sensor Type	10080	0		(0 = None) ; 1 = Magnetic 3 = Data Link - Tailshaft	1	1

Annexure 6

Maintenance Monitor						
Maintenance Monitor Enable	20131	N		Y = Enable N = Disable	Y	Y
Maintenance Monitor Alert Percentage	10860	90	Min: 0 Max: 100	%	90	90
Maintenance Monitor Operating Mode	10230	0	0=Automatic ; 1=Time 2=Fuel ; 3=Distance	0 = Automatic ; 1 = Time 2 = Fuel ; 3 = Distance	3	3
Maintenance Monitor Time	10290				1000 sec	1000 sec
Maintenance Monitor Distance	10280	25000	Min: 0 Max: 1000000	Miles	40000 km (24854.848 miles)	40000 km (24854.848 miles)
Maintenance Monitor Fuel	10250				2404	2404
Alternator Failure Warning	22571	N		Y = Enable N = Disable	Y	Y
Anti-Tampering Measures						
Maximum Engine Speed Without Vehicle Speed Source	10020	2850	Min: 900 Max: 3000	rpm	1800	1800
Vehicle Speed Sensor Anti-Tampering Enable	20761	N		Y = Enable N = Disable	Y	Y
Engine Protection						
Engine Protection Shutdown Enable	20091	Y		Y = Enable N = Disable	Y	Y
Service Brake						
Service Brake Switch Enable	50070	N		Y = Enable N = Disable	Y	Y
Fan Control						
Air Conditioning Pressure Switch Input	21712	Disable		Y = Enable N = Disable	Y	N
Min Fan ON Time for Air Conditioning Pressure Switch	11350			Seconds	5	---
Fan Type	12490				7	7
Fan drive ratio	50670			ratio	1.2	1.2
Air Condition Control (Applicable only for SIGNA cabin)						
Air Conditioner Control	51260				Y	N
Air Conditioner On Idle Speed Adjust	51262				Y	---
Low Speed Idle Setpoint with Air Conditioning	51264				900	---
Powertrain Protection (PTP)						
Powertrain Protection	21781	Disable		Y = Enable N = Disable	N	N

DEBABRATA DAS [TMCV, Engineering, Jamshedpur]

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune]
Sent: 08 March 2023 12:58
To: SWAPNIL JADHAV [TMCV, Engineering, Pune]; GURMEET SINGH [TMCV, Engineering, Pune]; ANIRUDDHA TIRUKHE [TMCV, Engineering, Pune]; SANDIP PHAPALE [TMCV, Engineering, Pune]; DEBABRATA DAS [TMCV, Engineering, Jamshedpur]
Subject: M&HCV Summer Trials

Dear Team,

PFA trailing mail for summer trials are not required for 5.6 and 6.7L engine.

Regards,

Ajay

From: AJAYKUMAR JADHAV [TMCV, Engineering, Pune]

Sent: 08 June 2022 17:18

To: 'Raj Mohanty' <raj.mohanty@cummins.com>; UDAY HADAP [ERC, Engineering, Pune] <uday.hadap@tatamotors.com>; SUBHAYU DAS [TMCV, Engineering, Jamshedpur] <subhayu.das@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>

Cc: Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Madhurima Biswas <madhurima.biswas@cummins.com>; PRAHLADA ANAND [ERC, Engineering, Pune] <anand.p@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>; Gupta Deepak <gupta.deepak@cummins.com>; VISHNU DIXIT [TMCV, Engineering, Jamshedpur] <vishnud@tatamotors.com>; SANDIP GAIDHANE [TMCV, Engineering, Pune] <sandip.gaidhane@tatamotors.com>; MAHESH PISE [TMCV, Engineering, Pune] <mahesh.pise@tatamotors.com>; VINODKUMAR S M [TMCV, Engineering, Pune] <vinodkumar.sm@tatamotors.com>; Rahul Singh (TTL) <rahuls4.ttl@tatamotors.com>

Subject: RE: M&HCV Summer Trials

Dear All,

As per trailing mail confirmation from Cummins team. Only LPC trials is planned on 4830.TK as worst case at JSR and summer trials are not required for 5.6 and 6.7L engine.

Subhayu/ Sanjeev Sir: Kindly confirm 48T vehicle availability for LPC trials at JSR . Cummins team will join trials along with instrument setup.

Thanks for confirmation.

Regards,

Ajay

From: Raj Mohanty <raj.mohanty@cummins.com>

Sent: 02 June 2022 15:58

To: UDAY HADAP [ERC, Engineering, Pune] <uday.hadap@tatamotors.com>

Cc: Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Madhurima Biswas <madhurima.biswas@cummins.com>; PRAHLADA ANAND [ERC, Engineering, Pune] <anand.p@tatamotors.com>; SANJEEV KUMAR [TMCV, Engineering, Jamshedpur] <sanjeev1.kumar@tatamotors.com>; AJAYKUMAR JADHAV [TMCV, Engineering, Pune] <ajaykumar_jadhav@tatamotors.com>; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>; Gupta Deepak <gupta.deepak@cummins.com>

Subject: RE: M&HCV Summer Trials

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Dear Uday,

LPC trial is planned in Signa 4830.TK vehicle at JSR as a worst case based on higher fuel layout.

Regards,

Raj Mohanty

Application Engineering

+91 90310 56223

From: uday.hadap@tatamotors.com <uday.hadap@tatamotors.com>

Sent: 30 May 2022 15:07

To: Gupta Deepak <gupta.deepak@cummins.com>

Cc: anand.p@tatamotors.com; sanjeev1.kumar@tatamotors.com; ajaykumar_jadhav@tatamotors.com; Abhijit M Chittewan <abhijit.chittewan@cummins.com>; G. Karthikeyan <karthikeyan.ganesan@cummins.com>; Sachin Solanki <sachin.solanki@cummins.com>

Subject: M&HCV Summer Trials

Importance: High

EXTERNAL SENDER: This email originated outside of Cummins. Do not click links or open attachments unless you verify the sender and know the content is safe.

Dear Deepak,

Reference discussion last week. Kindly confirm Cummins team availability for summer trials (Jaisalmer) with instrument & include the tests to be conducted.

Engine	Model	Responsible person	Location	Remark	Cummins team remark if any
5.6L	Signa 3023.T	Uday Hadap	Jaisalmer	Discussion under progress (Details will be shared shortly by Engine) Note: LPC trials not required for summer trials . final confirmation required from engine if any additional tests with supplier	
6.7L	Signa 5530.S	Uday Hadap	Jaisalmer		

Regards,

Uday

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